

GOLDILOCKS RADIX

What Graph Theory Tells Us About Radix, Degree & Valency

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DÆDÆLUS Research September 6, 2025



CENTRALIZED THINKING

Centralization leads not only to unilateral power of one group’s or individual’s control, it also leads to significant brittleness, as we’ve seen in the huge amplifications caused by outages at Akamai, Fastly & Cloudflare.

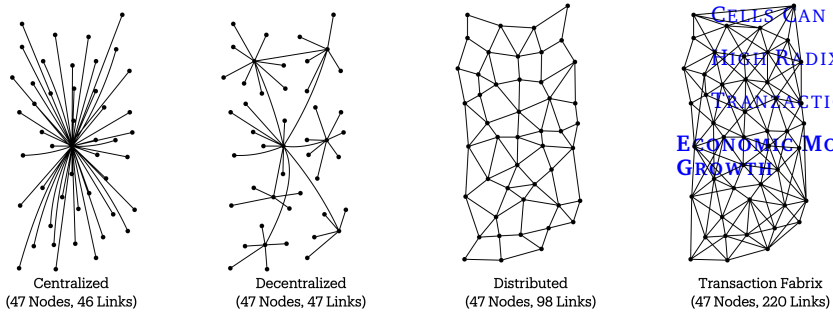


Figure 1: Baran: selection from “On Distributed Communications.”

BARAN CENTRALIZED

In the beginning, networks were physically, logically and virtually centralized. The argument is that it’s simpler to design and manage. But this was only true when one person was in charge of everything. It also had the convenient benefit (to that one person) that they gained power as more people are added to their network and were willing to pay (trade) to use their resource.

Ethernet started as a single broadcast domain coaxial cable where anyone could transmit and anyone can receive. It didn’t take long before Bob Metcalfe at 3COM realized they needed a “repeater” architecture, and that broadcast domains need to be segmented, so they could scale.

The centralized model is Single Point of Failure and Control (SPoFaC). It also represents a bottleneck for Any to Any (A2A) traffic trying to communicate through the centralized resource, by subjecting all users to the statistical whims of crowd behavior and congestion.

BARAN DECENTRALIZED

We learned how to make centralized resources (such as circuit switching in telephone systems) more efficient by introducing packet switching, so the same “circuits” could multiplex more users on the same resource. As network speeds increased, the internet opened up and we have the revolution today in the web, video and highly interac-

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tive social media. The Centralized physical model gave way to decentralized, and then to (almost) distributed. The IP Hourglass provided a stable interface where innovation could take place in the layers both above and below, and the modern internet prosperity ensued.

BARAN DISTRIBUTED

Paul Baran provided extensive mathematical rationale if not proofs for his design choices [[On Distributed Systems](#)]. These continue to provide insight as we think independently about physical, logical, and virtual abstractions. While this stratification can be extended indefinitely, we can illuminate using just three. The physical interconnect (Engineering: hardware, protocols, signaling), the logical connectivity (Resource, topology, provisioning, monitoring). And Virtual (Management: Ownership, Control and the [Extraction of rents](#) — which is [what Cloud has done](#)).

All three layers must independently trade off Paul Baran's spectrum of centralized, decentralized distributed (and the SPoFaC, bottlenecks and economic control, inherent, and separately disrupt-able in each layer, and on upwards into the stratification that combines our world economy.

THE TRANSACTION FABRIX

A direct follow-on to Paul Baran's reasoning about the Distributed architecture. Modern Graph Theory characterizes graph resources as vertices (nodes), and edges (links). There is a third perspective in planar arrays of computing elements: Degree (Graph Theory), Radix (Network Vendors), and Valency (Chiplet packaging).

If we imagine (for a minute) the desirability of homogeneous systems, which can readily exploit modern semiconductor manufacturing techniques, which have taken us from 2D arrays of transistors, to 2.5D and now a fully 3D structure which mimics modern skyscraper principles.

THE GOLDILOCKS DEGREE

All nodes with a single port are restricted to becoming leaves of the tree, which makes it necessary to have a separate switching network. If we need redundancy, we not only need more ports on our servers, we need fully redundant “planes” of network switching to go along with them.

At 2 ports, Baran's minimum redundancy factor $R=1$. At $R=2$ we have 4 ports. $R=4$ we have 8 ports, and can create a richly connected interconnect, which has many more paths to connect between any two points.

Using Graph Theory can from any node, count the number of alternate trees (for which it is the root) using Kirchhoff's Matrix Tree algorithm. By sending request/response message exchanges on its ports, so it can build its knowledge (information model) of its environment. Unlike the “planes”

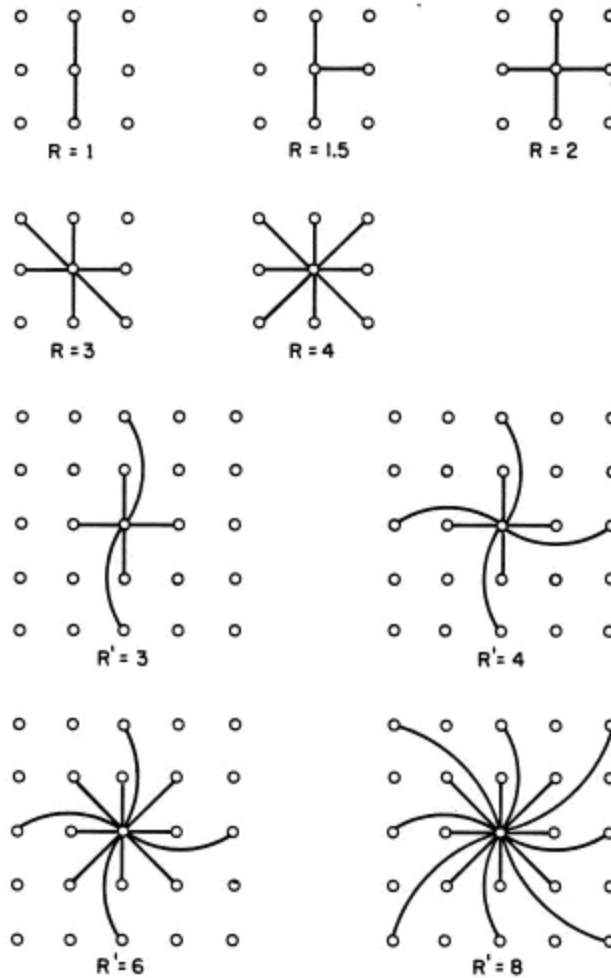


Figure 2: Embedded figure: pasted-image-29.

in conventional redundant Clos Networks, each port tree can use any path through the network to ensure it can communicate with its peers.

As we go up the network port count, we run into the law of diminishing returns. Resiliency of the graph continues to grow, but after a certain number of ports, the cost and complexity rise faster than the benefit, and we've hit an economic maximum.

To understand this new perspective, and make the results simple and obvious, we need to model different cost factors (as was done in [HammingMesh](#)).

Firstly, all links are not created equal. So we count lanes instead. Fatter (higher bandwidth) links are created by ganging together 2, 4, or 8 lanes. E.g., iGolf we use the (about to be commodity) 25Gbit lanes as a baseline, we can make 50, 100 and 200 Gbit links. To a first approximation, these cost 2, 4 and 8 times as much respectively.

Secondly, links between servers and switches, and between layers of switches, must necessarily be longer. Not only are longer cables more expensive, they trigger signal integrity and reliability thresholds where more energy is needed to drive longer cables, and justify the switch to optical.

Thirdly, optical cables are more expensive than copper cables. Typically twice the cost per link, although manufacturers will often try to artificially increase their margins by doing proprietary things in the implementations and claim their competitors are inferior. This can make the prices of these cables arbitrarily high, and rule becomes “what the market can bear”.

It doesn’t stop there. Packets that travel over a single link cost “one link’s worth of bandwidth”. Packets that travel over 8 or 10 links cost “8 or 10 links worth of bandwidth”. By forcing all the packets to go through the switched networks, and ignore locality in workload placement, we are ignoring an important economic tradeoff.

THE TROUBLE WITH LINK BONDING

Network folks like to measure everything as bandwidth. They often conflate link bandwidth and lane bandwidth by bonding multiple lanes (2, 4, 8 +) so that links can be marketed as having 2x, 4x or 8x the throughput. Throughput is easy for sales people to sell. More is better, right? In contrast, Latency, which is far more significant in its impact to distributed applications like databases and consensus protocols, is technically harder to achieve, but is a lot harder to sell. Switch vendors are selling bandwidth. What distributed systems developers want is interaction speed, not byte stream flows.

CELLS CAN DO FORWARDING

The Transaction Fabrix local topology was inspired by Paul Baran’s definition of redundancy. One hop valency takes us to 8 compass points (this is not a Torus addressing scheme). Two hop redundancy takes us out to a 5x5 tile, which, like the shift from circuit to packet switching, can provide a “next level” of communication using an even simpler redundancy scheme (cut through routing) which eliminates the need for store and forward buffers.

Network protocols (both Ethernet and IP), are fundamentally insecure. We’ve invested billions of dollars throughout the industry trying to “tack on” security in the form of IPSEC, ssh, PKI and any other band-aid’s that we can no longer rip off.

SDN was introduced to express a security predicate which is way to state, but difficult to implement. Or so the conventional wisdom goes.

HIGH RADIX THINKING

Coordination Patterns and Traffic

Like a city, you need a town hall, opera house and Stadium sometimes; but most of the time the traffic is local, and you need streets. By paying attention to the locality of the traffic patterns, and explicitly managing topology deterministically from the applications, instead of statistically from a black box network that forces the E2E principle on you, we can dramatically improve the economic model of networks, as can be seen in [HammingMesh](#).

Conventional Networks assume applications need a flat network (all nodes are equally distant from each other through a flat hierarchy

TRANZATIONS NOT FLOWS.

Larry Peterson says the internet seems to be missing a standard protocol for the request/response paradigm [[TCP vs. RPC all over again](#)].

We agree in principle: loose, *temporally disconnected* packets are a menace in distributed systems. There is no common knowledge about the transfer of information (what physicists call signaling), without request/reply message pairs.

The physics is clear [[2022 Nobel](#)] non-local correlations occur “instantaneously” over any arbitrary distance, but signaling always requires the speed of light. Spacetime, it appears, has a handshake.

Where we disagree, is that this can be reliably achieved using “logical channels” over existing networks. AWS’ (and all cloud vendors) blindness, is their insistence on the End To End ([E2E](#)) Principle, which they claim gives them the right to spray packets (using ECMP or EQDS), across multiple paths on the network, in the hope that the endpoint receivers can sort out the mess at the other end.

This bang bang bang ([imposition](#)) style networking continues to thwart distributed systems developers, who need messages to be *reliably paired*, not randomly dropped, reordered, duplicated or delayed.

The misunderstanding comes from the implicit and most often unstated assumption of Newtonian time underneath most algorithms involving distributed systems, network protocols and software coordination.

Byte streams bite

This is how Pat Helland describes this phenomenon. When you rely on the receiver to have an infinite amount of buffer space, you will run into problems. SmartNICs don’t have infinite resources. Unlike CPU’s which can evict cache lines to make more fast space available, or page out least recently used data to make buffers bigger, NICs have to treat packets like hot potatoes, they must be passed on as fast as possible, or dropped ...

Its Even worse for GC languages. Stop the world they say, we need to wash the dishes and take out the trash.

Politics, Warfare and Congestion

Napoleon won because he knew how to organize, assemble and plan. This was a triumph of centralization, and he filled a gap in human history that his fragmented opponents had no

hope of defending against because they could not innovate as fast as the smaller decentralized armies. [[Why Nobody Could Defeat Napoleon](#)].

At least for a while — until it works no more.

This cycle of *centralized*, *decentralized* and back to *distributed* occurs not only in datacenters, and the internet; It occurs in countries, warfare, international trade and politics. There is no single scheme that fits all. It's the cycling that creates the dynamics of an ever growing economy, complete with collapses and creative destruction.

ECONOMIC MODEL FOR GROWTH

When they are forced to, network engineers will distinguish between throughput (raw bandwidth) and goodput (useful bandwidth), but they do so from a concept of bisection bandwidth, without distinguishing between packets that are computationally useful, and those that are merely cloned as they transit the network,. In a directly connected mesh network,

Observer Theory

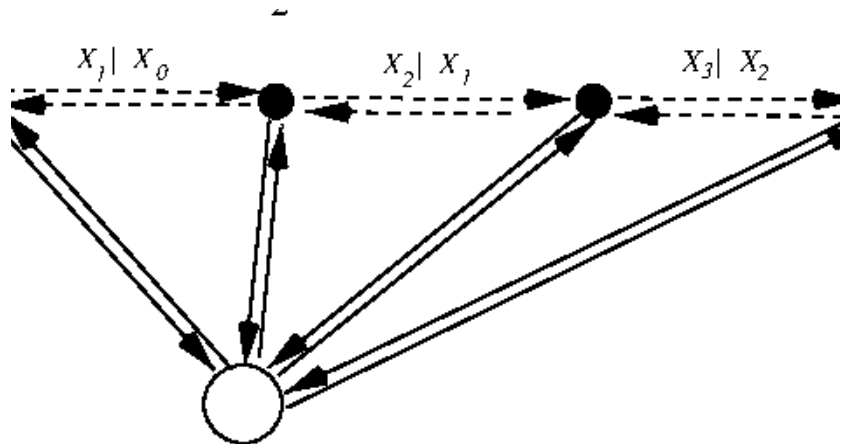


Figure 3: Embedded figure: pasted-image-33 (cropped).

From [Promise Theory](#).